# Bragg band gaps in a LEGO sonic crystal along $\Gamma X$ and $\Gamma M$

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**Abstract.** A sonic crystal blocks sound in a band of frequencies, the band gap, expected near $f_B = c/(2d)$, with $d$ the spacing between planes of scatterers. We measured the transmission of white noise (spectrum with crystal divided by spectrum without crystal) through square lattices of $2 \times 2$ LEGO columns, with lattice constant $a = 32$–$56$ mm and $N = 1$–$11$ rows, along $\Gamma X$ and $\Gamma M$ (lattice rotated by 45°), and compared it with a plane-wave expansion (PWE). Along $\Gamma X$ the lower gap edge agrees with the PWE within $1.3\sigma$ for $a \leq 48$ mm but not for $a = 56$ mm ($3.9\sigma$); the gap does not scale as $1/a$ ($\chi^2/\nu = 15$), and the attenuation grows linearly with $N$, by $1.45 \pm 0.29$ to $1.98 \pm 0.41$ dB per row ($a = 40$–$56$ mm). Rotating the crystal to $\Gamma M$ raises the lower edge by a factor 1.27–1.45, consistent with $\sqrt{2}$ within $2\sigma$. The precision is limited by the reference measurement and by sound bypassing the crystal, and we find no evidence of a complete gap.

## 1 Introduction

A sonic crystal is a periodic array of rigid obstacles in air. Reflections from successive planes of obstacles, separated by $d$, add in phase when $\lambda = 2d$, and around

$$f_B = \frac{c}{2d}, \qquad d = a \ (\Gamma X), \qquad d = a/\sqrt{2} \ (\Gamma M) \tag{1}$$

a band gap opens [1, 2]: at the edge of the Brillouin zone, $k = \pi/d$, the standing waves with maximum pressure on the planes of bricks and in the air between them no longer have the same frequency, and between the two, $f_{\text{inf}} < f_B < f_{\text{sup}}$, no wave propagates. Inside the gap $k = \pi/d + i\kappa$ and the pressure decays as $e^{-\kappa x}$; after $N$ rows the attenuation in decibels is

$$D(N) = \alpha N, \qquad \alpha = 20 \log_{10}(e)\, \kappa d. \tag{2}$$

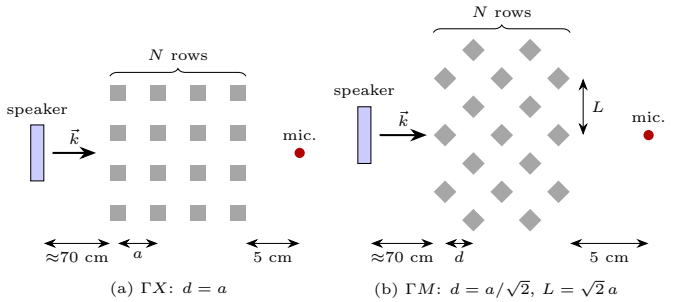
The position and width of the gap depend on the filling fraction $ff = s^2/a^2$ (0.24 to 0.08 here): the bricks slow the sound down, $c_{\text{eff}} \simeq c/\sqrt{1+ff}$ [4], and the gap moves below $f_B$, more so for smaller $a$. The bands are computed by solving $\nabla \cdot (\rho^{-1} \nabla p) + \omega^2 p/(\rho c^2) = 0$ with $p$ written as a Bloch wave (PWE, 225 plane waves, Appendix B). We tested, along $\Gamma X$ and $\Gamma M$, the gap edges, their dependence on $a$, the linear attenuation in $N$, and whether the gaps of the two directions overlap (complete gap).

## 2 Method

***Experimental set-up.*** The crystal is a square lattice of columns (Fig. 1), each one a stack of $2 \times 2$ LEGO bricks (side $s = 15.8$ mm) on a LEGO baseplate. The lattice constant $a$ is the centre-to-centre distance between neighbouring columns; since the studs of the baseplate are 8 mm apart, $a$ is a multiple of 8 mm.

Along $\Gamma X$ the rows are at $d = a$. Only this spacing along the propagation direction sets the Bragg frequency, $f_B = c/(2d)$, and it was the only one changed: the lateral spacing and the enclosure were kept fixed, so the entrance is always $W = 18$ studs (144 mm) wide. Along $\Gamma M$ the lattice is rotated by 45° (rows alternating between 4 and 3 bricks, with 3 at the entrance) and the two spacings can no longer be chosen separately: the planes are at $d = a/\sqrt{2}$ and the columns within a plane at $L = \sqrt{2}\, a = 2d$, so $f_B = c/L$. Increasing $a$ therefore widens the lattice laterally as well, and so the entrance, which holds three columns, grows with it: $W = 14$, 17 and 20 studs, i.e. 112, 136 and 160 mm, for $a = 32$, 40 and 48 mm.

The crystal is enclosed by LEGO walls and roof and placed in a cardboard box with foam around it. A Bluetooth loudspeaker $\approx 70$ cm from the crystal face emits white noise, with frequencies between 500 and 12 000 Hz, for 10 s, and a USB microphone 5 cm behind the crystal records the sound. The loudspeaker distance satisfies the plane-wave criterion (Appendix B).




(a) $\Gamma X$: $d = a$ (b) $\Gamma M$: $d = a/\sqrt{2}$, $L = \sqrt{2}\, a$

**Figure 1.** Set-up seen from above (not to scale). Each column is a stack of $2 \times 2$ LEGO bricks on a LEGO baseplate. (a) $\Gamma X$: aligned columns. (b) $\Gamma M$: lattice rotated by 45°, with a brick vertex facing the loudspeaker and rows alternating between 4 and 3 bricks. $a$: distance between columns $d$: distance between planes; $L$: period within a plane.

***Parameters varied and repetitions.*** We varied $a = 32$, 40, 48 and 56 mm along $\Gamma X$ and $a = 32$, 40 and 48 mm along $\Gamma M$, and the number of rows $N$, removing one row at a time from the microphone side ($N = 10 \to 1$ along $\Gamma X$ and $N = 11 \to 1$ along $\Gamma M$, except along $\Gamma M$ with $a = 32$ mm, where $N = 1$ was not measured). The loudspeaker, the microphone and the volume were kept fixed. Each configuration had 5 recordings, for a total of 72 configurations and 360 recordings, of which 16 were excluded for saturation or Bluetooth dropouts. The 4 sessions (1 and 6 October along $\Gamma X$, 12 and 13 October along $\Gamma M$) took place at $\theta = 20.5\,^{\circ}$C and $20.0\,^{\circ}$C ($\pm 1\,^{\circ}$C), which gives $c = (343.5 \pm 0.6)$ m/s and $(343.2 \pm 0.6)$ m/s.

***Quantity measured and data processing.*** We measured the relative transmission

$$T(f) = 10 \log_{10} \left[ \frac{\bar{P}_{\text{crystal}}(f)}{\bar{P}_{\text{ref}}(f)} \right] \ (\text{dB}), \tag{3}$$

where $P$ is the power spectral density of the recorded signal, analysed from 500 Hz to 12 kHz, and the reference is the same set-up without columns. The division removes the response of the loudspeaker, the microphone and the room.

*Spectra.* $P$ was computed with Welch's method (Hann window, 4096 points, $\Delta f = 11.7$ Hz (Frequency resolution of the spectrum) at a sampling rate of 48 kHz) on the segment 0.8–10.3 s of each recording, which excludes the Bluetooth delay.

*Smoothing.* To reduce the statistical fluctuations of the spectrum, which are amplified when dividing two spectra, each $P$ was smoothed in power, i.e. the value at each frequency $f$ was replaced by the mean power in a window around $f$ (and not by the mean in dB, to avoid biasing the result). The window is 1/6 of an octave wide (an octave is an interval in which the frequency doubles), i.e. it goes from $f/1.06$ to $f \cdot 1.06$ ($\pm 6\,\%$ of $f$).

*Reference.* The reference was measured for each value of $a$ at the start (and also at the end, along $\Gamma M$) of the measurements. For each $a$ we used, among the references of the session, the one that makes $T(f)$ flattest in the control band (see below); the others enter the uncertainty associated with the reference.

***Control band.*** As a check of the measurement, $T$ should be constant and close to 0 dB in the band 0.35–0.70 $f_0$, with $f_0 = c/(2a)$ along both directions. This band lies at frequencies below the first gap of the lattice, where the crystal is prac-

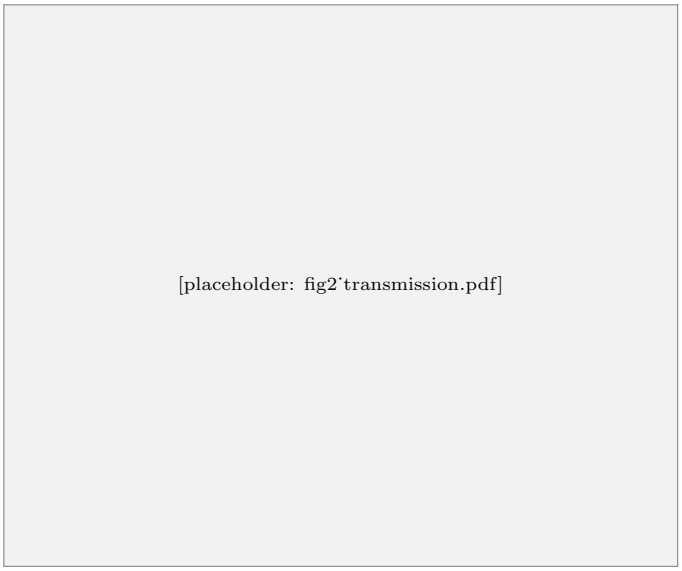



**Figure 2.** Relative transmission for $a = 48\,\text{mm}$. Grey: PWE gap; red: measured edges (mean over $N \geq 4$). Along $\Gamma X$ the gap closes; along $\Gamma M$ lateral diffraction starts at $f_B$ and $T$ does not return to 0. All $a$ in Appendix A.

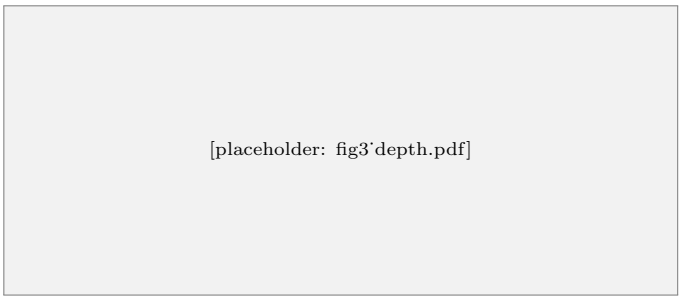



**Figure 3.** Depth $D$ at $f_B \pm 10\,\%$ versus number of rows, with the line $D = \alpha N + b$ fitted for each $a$ (slopes in Table 1).

tically transparent. Along $\Gamma M$ we use $f_0 = c/(2a)$ and not $f_B = c/(2d)$, because the band 0.35–0.70 $f_B$ would reach the $\Gamma X$ gap.

## 3 Analysis

From each $T(f)$ we extract: the base level $T_0$ (median in the control band 0.35–0.70 $c/(2a)$, transparent along both directions) and the *ripple* (robust deviation of $T$ in that band, where $T$ should be flat); the gap, a continuous region with $T < T_0 - 6\,\text{dB}$ that touches 0.75–1.25 $f_B$, whose threshold crossings are $f_{\text{inf}}$ and $f_{\text{sup}}$ (the gap "closes" if $f_{\text{sup}} < 1.5 f_B$); and the depth $D = T_0 - \langle T \rangle_{f_B \pm 10\%}$. The windows are in units of $f_B$, so the calculations are the same along both directions.

*Uncertainties.* For each number, $\sigma^2 = \sigma^2_{\text{est}} + \sigma^2_{\text{rip}} + \sigma^2_{\text{ref}}$ ($+\sigma^2_{\text{res}}$): dispersion of the 5 repetitions ($/\sqrt{5}$, almost always negligible), propagated ripple (0.92×ripple in $D$; ripple/$|dT/df|$ at the edges), dispersion among the references of the same day, and $\Delta f/\sqrt{12} = 3.4\,\text{Hz}$. Per block, the frequencies are the weighted mean over $N \geq 4$, with $\sigma^2 = \sigma^2_{\text{disp}} + \sigma^2_{\text{sys}}$: the systematic part (ripple and reference, common to the whole block) does not decrease with averaging. Fits are weighted least squares ($\sigma_a = 0.1\,\text{mm}$ through effective variance); if $\chi^2/\nu > 1$ the uncertainties are multiplied by $\sqrt{\chi^2/\nu}$. The theory has an uncertainty from the brick side ($\pm 0.1\,\text{mm}$) and from the PWE truncation (0.5 %). Comparisons use $z = |x - x_0|/\sqrt{\sigma_x^2 + \sigma_0^2}$.

## 4 Results

*Gap edges.* Along $\Gamma X$ there is a well-defined gap for $N \geq 4$ for all $a$ (Fig. 2); it closes in 27 of the 28 measurements with $N \geq 4$. The edges are in Table 1: for $a \leq 48\,\text{mm}$, $f_{\text{inf}}$ agrees with the PWE within $1.3\sigma$ and $f_{\text{sup}}$ within $2.6\sigma$; for $a = 56\,\text{mm}$ both lie above ($z = 3.9$ and 3.6).

*Dependence on a.* The line $f_c = m/(2a)$ through the origin gives $m = (339 \pm 15)\,\text{m/s}$, compatible with $c = (343.5 \pm 0.6)\,\text{m/s}$ ($z = 0.3$), but with $\chi^2/\nu = 15.4$ ($p = 5 \times 10^{-10}$): on average the centre is at $f_B$, but it is not proportional to $1/a$. The free line describes the points ($\chi^2/\nu = 1.4$) with $m = (229 \pm 20)\,\text{m/s}$ and intercept $(1283 \pm 231)\,\text{Hz}$. For $f_{\text{inf}}$ the same holds: through the origin $m = (278 \pm 20)\,\text{m/s}$ with $\chi^2/\nu = 30$ (Appendix A). The PWE gives $f_c/f_B = 0.925$–$0.968$.

*Attenuation.* Along $\Gamma X$ the depth grows linearly with $N$ (Fig. 3): for $a = 40$, 48 and 56 mm the intercept is $b = -0.2 \pm 2.0$, $-0.6 \pm 1.9$ and $-0.3 \pm 0.6\,\text{dB}$ (compatible with 0, as Eq. (2) predicts) and $\chi^2/\nu = 0.3$–$1.4$. For $a = 32\,\text{mm}$, $D$ saturates at $\approx 13\,\text{dB}$ from $N = 4$ and the line gives $b = 5.1 \pm 2.9\,\text{dB}$. The measured $\alpha$ is 15, 50, 53 and 80 % of the PWE value for $a = 32$–$56\,\text{mm}$.

*Direction $\Gamma M$.* $f_{\text{inf}}$ is 8–11 % below the PWE ($z = 2.5$–$3.6$) and the gap closes in only 3/8, 5/8 and 4/8 of the measurements ($a = 32$, 40, 48 mm). The ratio $f_{\text{inf}}(\Gamma M)/f_{\text{inf}}(\Gamma X)$ is 1.445(78), 1.271(72) and 1.360(56), versus $\sqrt{2}$ (Bragg; $z = 0.4$, 2.0, 1.0) and 1.59, 1.51, 1.47 (PWE; $z = 1.8$, 3.3, 1.9). The measured overlap $f_{\text{sup}}(\Gamma X) - f_{\text{inf}}(\Gamma M)$ is $423 \pm 331$, $560 \pm 286$ and $191 \pm 163\,\text{Hz}$, less than $2\sigma$ above zero; the PWE gives $-143$, $-357$ and $-505\,\text{Hz}$ (no complete gap).

**Table 1.** Results per block (mean over $N \geq 4$) versus the PWE. Frequencies in Hz; $\alpha$ in dB/row. † The gap does not close. ‡ Only 14–38 % of the $f_B \pm 10\,\%$ window lies inside the theoretical gap: compares orders of magnitude only.

| | $a$ | $f_B$ | $f_{\text{inf}}$ | | | $f_{\text{sup}}$ | | | $\alpha$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| | mm | | meas. | PWE | $z$ | meas. | PWE | $z$ | meas. | PWE |
| $\Gamma X$ | 32 | 5367 | 3806(84) | 3892 | 1.0 | 5923(191) | 6037 | 0.6 | 0.81(39) | 5.4 |
| | 40 | 4294 | 3586(151) | 3387 | 1.3 | 5117(227) | 4752 | 1.6 | 1.98(41) | 4.0 |
| | 48 | 3578 | 2969(102) | 2988 | 0.2 | 4230(135) | 3880 | 2.6 | 1.45(29) | 2.8 |
| | 56 | 3067 | 2917(62) | 2666 | 3.9 | 3722(124) | 3271 | 3.6 | 1.53(11) | 1.9 |
| $\Gamma M$ | 32 | 7584 | 5500(271) | 6180 | 2.5 | —† | 6990 | — | 0.73(20) | 0.1‡ |
| | 40 | 6067 | 4557(173) | 5109 | 3.1 | 7149(410) | 5818 | 3.2 | 0.81(28) | 0.5‡ |
| | 48 | 5056 | 4040(93) | 4385 | 3.6 | —† | 4942 | — | 1.30(17) | 0.6‡ |

## 5 Discussion

$\Gamma X$. A PWE for an infinite 2D crystal of rigid rods describes the lower edge for $a \leq 48\,\text{mm}$ without free parameters, and the gap below $f_B$ for $a = 32\,\text{mm}$ ($ff = 0.24$) is consistent with the slowing of sound by the bricks. This is what makes the line through the origin fail: the slope $(339 \pm 15)\,\text{m/s}$ would be "the speed of sound" if the gap were always at $f_B$, but the $\chi^2/\nu$ shows that it is not. The measured trend in $f_c/f_B$ is stronger than the PWE one, especially for $a = 56\,\text{mm}$, where the theoretical gap is only 600 Hz wide, comparable to the 1/6-octave smoothing ($\approx 350\,\text{Hz}$ at 3 kHz), and the crossings of a fixed threshold no longer coincide with the band edges. The $a = 40\,\text{mm}$ block mixed two sessions, hence the larger systematic uncertainty.

*Attenuation.* $D \propto N$ with $b \approx 0$ is the signature of the evanescent Bloch wave [3]. But the crystal is only 3.8 cm tall and 11–16 cm wide: some sound goes over the top and around the sides, which imposes a ceiling of ≈13–15 dB. The larger the theoretical $\alpha$, the sooner the ceiling is reached: hence the measured/theory fraction grows from 15 % ($a = 32$) to 80 % ($a = 56$).

$\Gamma M$. With the source at $\sim 70\,\text{cm}$ and a narrow crystal, sound that crosses the crystal obliquely, close to $\Gamma X$, whose gap is lower, reaches the microphone: $T$ starts to drop before the $\Gamma M$ gap and $f_{\text{inf}}$ comes out low. Above $f_B$ the rows, of period $L = \sqrt{2}a$, diffract sound sideways from $c/L = f_B$ and the sound no longer reaches the on-axis microphone; the gap stays "open". For the same reasons the measured overlap does not prove a complete gap.

*Limitations.* The reference dominates the error: moving the loudspeaker or the microphone by 1 cm changes the spectrum by several dB, and the ripple of $T$ (0.5–8 dB) is much larger than the dispersion of the repetitions. A rigid lid, a wider crystal and repeated reassemblies would reduce and measure these effects.

## 6 Conclusion

A LEGO crystal with $a = 32$–$56\,\mathrm{mm}$ has a Bragg gap along $\Gamma X$ whose lower edge agrees with the PWE within $1.3\sigma$ for $a \leq 48\,\mathrm{mm}$, and which does not scale as $1/a$ because of the filling fraction. The attenuation is linear in $N$, $\alpha \approx 1.5$–$2\,\mathrm{dB/row}$, limited to $\approx 15\,\mathrm{dB}$ by sound that goes around the crystal. Along $\Gamma M$ the gap appears about $\sqrt{2}$ higher, but the lower edge is 8–11 % below the PWE and the gap does not close, because of oblique propagation and lateral diffraction; there is no evidence of a complete gap.

## References


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## A Additional data and measurements

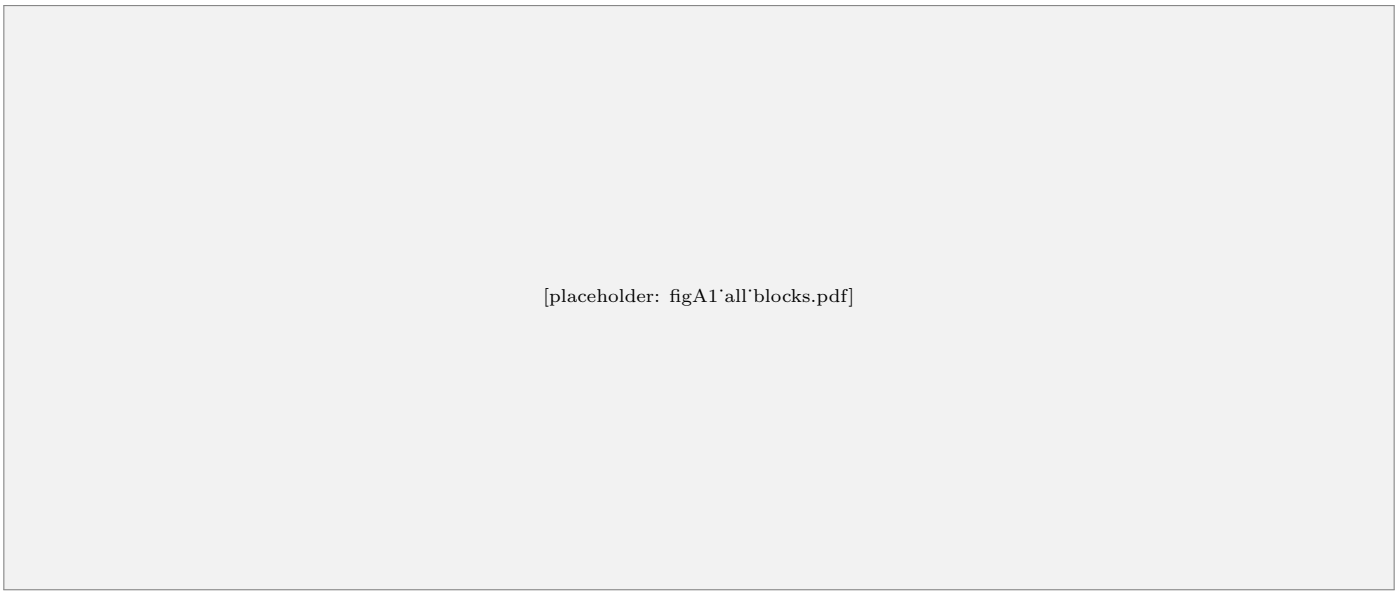



**Figure 4.** $T - T_0$ for all blocks with $N \geq 4$ (top: $\Gamma X$; bottom: $\Gamma M$). Band: PWE gap; red dashed: onset of lateral diffraction, $c/L$; dotted: threshold $T_0 - 6\,\mathrm{dB}$. Along $\Gamma M$ there are no measurements with $a = 56\,\mathrm{mm}$ and $N \geq 4$.




**Figure 5.** $f_{\mathrm{inf}}$ and $f_c$ (mean over $N \geq 4$ for each $a$) versus $x = 1/(2d)$, with the line through the origin and the free line; below, normalised residuals. Dashed: $f = c\,x$.




**Figure 6.** Top: $\Gamma X$ ($N = 10$) and $\Gamma M$ ($N = 11$) for the same $a$, with the measured gaps (bands) and $f_B$ (dashed). Bottom: $f_{\rm inf}/f_B$ and $f_c/f_B$ along both directions.

**Table 2.** All measurements. "rip.": ripple of $T$ in the control band (dB); "$k/5$ rep.": repetitions used; "ref. other $a$": reference from the same session but from another block. Total uncertainty $\sigma^2 = \sigma^2_{\text{est}} + \sigma^2_{\text{rip}} + \sigma^2_{\text{ref}}$ ($+\sigma^2_{\text{res}}$). The components are also in `tabela.csv`.

| dir | $a$ (mm) | $N$ | day | reference | rip. | $D$ (dB) | $f_{\text{inf}}$ (Hz) | $f_{\text{sup}}$ (Hz) | notes |
|---|---|---|---|---|---|---|---|---|---|
| $\Gamma X$ | 32 | 1 | 10-01 | ref_a32 | 4.2 | $-1.3 \pm 4.3$ | — | — | |
| $\Gamma X$ | 32 | 2 | 10-01 | ref_a32 | 6.2 | $0.1 \pm 6.0$ | — | — | |
| $\Gamma X$ | 32 | 3 | 10-01 | ref_a32 | 7.6 | $3.5 \pm 7.2$ | $5529 \pm 889$ | $5609 \pm 599$ | |
| $\Gamma X$ | 32 | 4 | 10-01 | ref_a32 | 1.7 | $11.4 \pm 2.1$ | $3947 \pm 195$ | $5768 \pm 1747$ | 4/5 rep. |
| $\Gamma X$ | 32 | 5 | 10-01 | ref_a32 | 1.8 | $8.0 \pm 2.7$ | $3846 \pm 133$ | $5622 \pm 161$ | |
| $\Gamma X$ | 32 | 6 | 10-01 | ref_a32 | 2.2 | $12.7 \pm 2.8$ | $3795 \pm 67$ | $5802 \pm 109$ | |
| $\Gamma X$ | 32 | 7 | 10-01 | ref_a32 | 1.7 | $13.5 \pm 2.3$ | $3781 \pm 70$ | $6431 \pm 165$ | |
| $\Gamma X$ | 32 | 8 | 10-01 | ref_a32 | 2.1 | $13.4 \pm 2.8$ | $3880 \pm 91$ | $6028 \pm 181$ | |
| $\Gamma X$ | 32 | 9 | 10-01 | ref_a32 | 1.5 | $9.9 \pm 1.8$ | $3794 \pm 79$ | $5981 \pm 893$ | |
| $\Gamma X$ | 32 | 10 | 10-01 | ref_a32 | 1.6 | $13.1 \pm 2.0$ | $3773 \pm 78$ | $5890 \pm 919$ | 4/5 rep. |
| $\Gamma X$ | 40 | 1 | 10-06 | ref_a56 | 3.1 | $-4.2 \pm 3.1$ | — | — | 4/5 rep., ref. other $a$ |
| $\Gamma X$ | 40 | 2 | 10-06 | ref_a56 | 5.4 | $3.9 \pm 5.1$ | $4538 \pm 944$ | $4713 \pm 254$ | ref. other $a$ |
| $\Gamma X$ | 40 | 3 | 10-06 | ref_a56 | 1.0 | $6.0 \pm 1.2$ | $3196 \pm 330$ | $4832 \pm 126$ | 4/5 rep., ref. other $a$ |
| $\Gamma X$ | 40 | 4 | 10-06 | ref_a56 | 1.2 | $8.9 \pm 1.6$ | $3474 \pm 168$ | $5677 \pm 114$ | ref. other $a$ |
| $\Gamma X$ | 40 | 5 | 10-06 | ref_a56 | 4.2 | $7.2 \pm 4.1$ | $3625 \pm 327$ | $4783 \pm 163$ | ref. other $a$ |
| $\Gamma X$ | 40 | 6 | 10-06 | ref_a56 | 2.0 | $14.5 \pm 2.0$ | $3525 \pm 122$ | $4818 \pm 271$ | ref. other $a$ |
| $\Gamma X$ | 40 | 7 | 10-01 | ref_a40 | 4.4 | $19.0 \pm 5.3$ | $3696 \pm 157$ | $4875 \pm 185$ | |
| $\Gamma X$ | 40 | 8 | 10-01 | ref_a40 | 4.7 | $10.1 \pm 4.8$ | $3833 \pm 213$ | $5510 \pm 462$ | |
| $\Gamma X$ | 40 | 9 | 10-01 | ref_a40 | 1.9 | $13.2 \pm 3.2$ | $3669 \pm 330$ | $4854 \pm 111$ | |
| $\Gamma X$ | 40 | 10 | 10-01 | ref_a40 | 2.2 | $18.2 \pm 4.1$ | $3556 \pm 101$ | — | does not close |
| $\Gamma X$ | 48 | 1 | 10-06 | ref_a48 | 1.9 | $-1.1 \pm 2.7$ | — | — | |
| $\Gamma X$ | 48 | 2 | 10-06 | ref_a48 | 2.4 | $2.5 \pm 2.7$ | — | — | 4/5 rep. |
| $\Gamma X$ | 48 | 3 | 10-06 | ref_a48 | 2.3 | $4.6 \pm 2.7$ | $3600 \pm 502$ | $3770 \pm 1598$ | |
| $\Gamma X$ | 48 | 4 | 10-06 | ref_a48 | 2.3 | $5.2 \pm 2.9$ | $3613 \pm 568$ | $4173 \pm 364$ | |
| $\Gamma X$ | 48 | 5 | 10-06 | ref_a48 | 2.7 | $8.7 \pm 2.8$ | $2883 \pm 125$ | $4157 \pm 260$ | |
| $\Gamma X$ | 48 | 6 | 10-06 | ref_a48 | 2.5 | $9.6 \pm 2.9$ | $2970 \pm 100$ | $4071 \pm 161$ | 4/5 rep. |
| $\Gamma X$ | 48 | 7 | 10-06 | ref_a48 | 2.0 | $9.4 \pm 2.6$ | $2982 \pm 98$ | $4093 \pm 130$ | 4/5 rep. |
| $\Gamma X$ | 48 | 8 | 10-06 | ref_a48 | 2.0 | $10.7 \pm 2.1$ | $2974 \pm 83$ | $4306 \pm 112$ | |
| $\Gamma X$ | 48 | 9 | 10-06 | ref_a48 | 2.3 | $9.6 \pm 2.8$ | $3019 \pm 79$ | $4332 \pm 96$ | |
| $\Gamma X$ | 48 | 10 | 10-06 | ref_a48 | 2.8 | $15.0 \pm 2.7$ | $2850 \pm 135$ | $4210 \pm 123$ | |
| $\Gamma X$ | 56 | 1 | 10-06 | ref_a56 | 0.8 | $0.7 \pm 0.7$ | — | — | |
| $\Gamma X$ | 56 | 2 | 10-06 | ref_a56 | 1.1 | $2.7 \pm 1.0$ | — | — | 4/5 rep. |
| $\Gamma X$ | 56 | 3 | 10-06 | ref_a56 | 0.5 | $4.4 \pm 0.5$ | $3018 \pm 50$ | $3142 \pm 230$ | |
| $\Gamma X$ | 56 | 4 | 10-06 | ref_a56 | 1.1 | $5.7 \pm 1.0$ | $2993 \pm 75$ | $3418 \pm 200$ | |
| $\Gamma X$ | 56 | 5 | 10-06 | ref_a56 | 1.6 | $9.1 \pm 1.5$ | $2912 \pm 72$ | $3607 \pm 133$ | 4/5 rep. |
| $\Gamma X$ | 56 | 6 | 10-06 | ref_a56 | 1.6 | $10.6 \pm 1.5$ | $2941 \pm 56$ | $3587 \pm 98$ | 4/5 rep. |
| $\Gamma X$ | 56 | 7 | 10-06 | ref_a56 | 0.8 | $11.3 \pm 0.7$ | $2884 \pm 46$ | $3593 \pm 77$ | |
| $\Gamma X$ | 56 | 8 | 10-06 | ref_a56 | 1.3 | $11.0 \pm 1.2$ | $2979 \pm 47$ | $3993 \pm 103$ | |
| $\Gamma X$ | 56 | 9 | 10-06 | ref_a56 | 1.5 | $15.1 \pm 1.5$ | $2595 \pm 266$ | $3809 \pm 215$ | |
| $\Gamma X$ | 56 | 10 | 10-06 | ref_a56 | 1.0 | $13.3 \pm 0.9$ | $2821 \pm 58$ | $3886 \pm 85$ | |
| $\Gamma M$ | 32 | 2 | 10-13 | ref_a32 | 1.6 | $3.3 \pm 1.5$ | $6288 \pm 297$ | $6883 \pm 150$ | |
| $\Gamma M$ | 32 | 3 | 10-13 | ref_a32 | 1.7 | $4.0 \pm 1.6$ | $5950 \pm 96$ | $6953 \pm 128$ | |
| $\Gamma M$ | 32 | 4 | 10-13 | ref_a32 | 2.0 | $3.9 \pm 1.8$ | $7871 \pm 1018$ | $8223 \pm 817$ | 4/5 rep. |
| $\Gamma M$ | 32 | 5 | 10-13 | ref_a32 | 2.0 | $2.7 \pm 1.9$ | $6005 \pm 2114$ | $6989 \pm 120$ | |
| $\Gamma M$ | 32 | 6 | 10-13 | ref_a32 | 2.0 | $7.4 \pm 1.9$ | $6076 \pm 179$ | — | does not close |
| $\Gamma M$ | 32 | 7 | 10-13 | ref_a32 | 1.8 | $8.4 \pm 1.6$ | $5910 \pm 73$ | — | 4/5 rep., does not close |
| $\Gamma M$ | 32 | 8 | 10-13 | ref_a32 | 1.6 | $10.0 \pm 1.5$ | $4603 \pm 189$ | — | does not close |
| $\Gamma M$ | 32 | 9 | 10-13 | ref_a32 | 1.9 | $5.7 \pm 1.8$ | $4680 \pm 206$ | $8281 \pm 309$ | |
| $\Gamma M$ | 32 | 10 | 10-13 | ref_a32 | 1.9 | $8.7 \pm 1.8$ | $4581 \pm 141$ | — | does not close |
| $\Gamma M$ | 32 | 11 | 10-13 | ref_a32 | 2.4 | $9.2 \pm 2.2$ | $4664 \pm 353$ | — | does not close |
| $\Gamma M$ | 40 | 1 | 10-13 | ref_a40_fim | 5.3 | $-2.1 \pm 4.9$ | — | — | |
| $\Gamma M$ | 40 | 2 | 10-13 | ref_a40_fim | 1.4 | $4.8 \pm 1.3$ | $6260 \pm 302$ | $6794 \pm 1291$ | |
| $\Gamma M$ | 40 | 3 | 10-13 | ref_a40_fim | 1.9 | $6.1 \pm 1.7$ | $4712 \pm 335$ | $5756 \pm 593$ | 4/5 rep. |
| $\Gamma M$ | 40 | 4 | 10-13 | ref_a40_fim | 1.4 | $7.7 \pm 1.3$ | $4633 \pm 101$ | — | 4/5 rep., does not close |
| $\Gamma M$ | 40 | 5 | 10-13 | ref_a40_fim | 1.9 | $10.2 \pm 1.7$ | $4695 \pm 445$ | $7151 \pm 209$ | |
| $\Gamma M$ | 40 | 6 | 10-13 | ref_a40_fim | 1.9 | $12.4 \pm 1.7$ | $4706 \pm 213$ | $8699 \pm 493$ | |
| $\Gamma M$ | 40 | 7 | 10-13 | ref_a40_fim | 2.3 | $11.4 \pm 2.2$ | $4535 \pm 210$ | $6865 \pm 1507$ | |
| $\Gamma M$ | 40 | 8 | 10-13 | ref_a40_fim | 2.0 | $14.1 \pm 1.8$ | $4544 \pm 175$ | $8655 \pm 429$ | |
| $\Gamma M$ | 40 | 9 | 10-13 | ref_a40_fim | 2.5 | $11.1 \pm 2.3$ | $4624 \pm 342$ | — | does not close |
| $\Gamma M$ | 40 | 10 | 10-13 | ref_a40_fim | 2.6 | $7.4 \pm 2.4$ | $4616 \pm 184$ | — | does not close |
| $\Gamma M$ | 40 | 11 | 10-13 | ref_a40_fim | 2.6 | $9.9 \pm 2.4$ | $4206 \pm 165$ | $6696 \pm 177$ | 4/5 rep. |
| $\Gamma M$ | 48 | 1 | 10-12 | ref_a48_fim | 1.4 | $0.4 \pm 1.5$ | — | — | |
| $\Gamma M$ | 48 | 2 | 10-12 | ref_a48_fim | 1.3 | $4.2 \pm 1.8$ | $5838 \pm 656$ | — | 4/5 rep. |
| $\Gamma M$ | 48 | 3 | 10-12 | ref_a48_fim | 1.2 | $6.5 \pm 1.7$ | $3896 \pm 112$ | — | |
| $\Gamma M$ | 48 | 4 | 10-12 | ref_a48_fim | 1.3 | $8.3 \pm 1.6$ | $4148 \pm 215$ | $6287 \pm 557$ | |
| $\Gamma M$ | 48 | 5 | 10-12 | ref_a48_fim | 1.9 | $8.5 \pm 2.3$ | $4131 \pm 176$ | $6232 \pm 769$ | |
| $\Gamma M$ | 48 | 6 | 10-12 | ref_a48_fim | 1.0 | $9.6 \pm 1.2$ | $3966 \pm 61$ | — | does not close |
| $\Gamma M$ | 48 | 7 | 10-12 | ref_a48_fim | 1.7 | $7.0 \pm 2.2$ | $4314 \pm 76$ | $5068 \pm 144$ | |
| $\Gamma M$ | 48 | 8 | 10-12 | ref_a48_fim | 2.7 | $8.7 \pm 2.7$ | $4360 \pm 236$ | $6941 \pm 973$ | |
| $\Gamma M$ | 48 | 9 | 10-12 | ref_a48_fim | 0.9 | $13.6 \pm 1.3$ | $3945 \pm 50$ | — | does not close |
| $\Gamma M$ | 48 | 10 | 10-12 | ref_a48_fim | 1.2 | $14.8 \pm 1.4$ | $3990 \pm 85$ | — | does not close |
| $\Gamma M$ | 48 | 11 | 10-12 | ref_a48_fim | 2.1 | $12.2 \pm 3.0$ | $4260 \pm 188$ | — | does not close |

## B Calibrations and derivations

***Speed of sound..*** $c(\theta) = 331.3\sqrt{1+\theta/273.15}\,\mathrm{m/s}$, $\sigma_c = c\,\sigma_\theta/[2(273.15+\theta)] = 0.6\,\mathrm{m/s}$ for $\sigma_\theta = 1\,^\circ\mathrm{C}$.

***Plane-wave Criterion..*** The loudspeaker emits approximately spherical waves. For the wave at the crystal entrance to be treated as plane, the distance $r$ from the loudspeaker to the entrance must be large compared with the entrance width $W$. The loudspeaker faces the centre of the entrance, so its edges are at $W/2$ from the axis and are reached after an extra path, relative to the centre, of $\delta = \sqrt{r^2+(W/2)^2} - r \approx W^2/(8r)$. This extra path corresponds to a phase difference of $360^\circ\,\delta/\lambda$ across the entrance, and we require $\delta \le \lambda/10$ (at most 36°), which gives

$$r \ge \frac{1.25\,W^2}{\lambda_{\min}}, \tag{4}$$

where $\lambda_{\min} = c/f_{\max}$ is the smallest wavelength used in the analysis: $\delta$ does not depend on frequency, but $\lambda$ decreases with it, so the condition is most demanding at the highest frequency. The highest frequency we use is the upper gap edge, which in Table 1 is at most $1.21\,f_B$ ($a = 56\,\mathrm{mm}$), so we take $f_{\max} = 1.2 f_B$, i.e. $\lambda_{\min} = 2d/1.2$. Along $\Gamma X$, $W = 18$ studs $= 144\,\mathrm{mm}$ for all $a$; along $\Gamma M$, $W = 14$, 17 and 20 studs (112, 136 and 160 mm) for $a = 32$, 40 and 48 mm.

**Table 3.** Minimum distance $r_{\min} = 1.25\,W^2/\lambda_{\min}$, with $\lambda_{\min} = 2d/1.2$ (upper edge of the gap), and margin with respect to $r = 70\,\mathrm{cm}$.

| Lattice | $a$ (mm) | $d$ (mm) | $W$ (mm) | $\lambda_{\min}$ (mm) | $r_{\min}$ (cm) | 70 cm/$r_{\min}$ |
|---|---|---|---|---|---|---|
| $\Gamma X$ | 32 | 32.0 | 144 | 53.3 | 48.6 | 1.44 |
| $\Gamma X$ | 40 | 40.0 | 144 | 66.7 | 38.9 | 1.80 |
| $\Gamma X$ | 48 | 48.0 | 144 | 80.0 | 32.4 | 2.16 |
| $\Gamma X$ | 56 | 56.0 | 144 | 93.3 | 27.8 | 2.52 |
| $\Gamma M$ | 32 | 22.6 | 112 | 37.7 | 41.6 | 1.68 |
| $\Gamma M$ | 40 | 28.3 | 136 | 47.1 | 49.0 | 1.43 |
| $\Gamma M$ | 48 | 33.9 | 160 | 56.6 | 56.6 | 1.24 |

In all configurations $r = 70\,\mathrm{cm}$ exceeds $r_{\min}$, with a minimum margin of 1.24 ($\Gamma M$, $a = 48\,\mathrm{mm}$).

***Plane-wave expansion (PWE)..*** With $\sigma = 1/\rho$, $\eta = 1/(\rho c^2)$ and $p(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\sum_{\mathbf{G}} p_{\mathbf{G}}\, e^{i\mathbf{G}\cdot\mathbf{r}}$, $\mathbf{G} = \frac{2\pi}{a}(m,n)$, the wave equation gives, for each $\mathbf{k}$, the eigenvalue problem

$$\sum_{\mathbf{G}'} \sigma(\mathbf{G}-\mathbf{G}')\,(\mathbf{k}+\mathbf{G})\cdot(\mathbf{k}+\mathbf{G}')\,p_{\mathbf{G}'} = \omega^2 \sum_{\mathbf{G}'} \eta(\mathbf{G}-\mathbf{G}')\,p_{\mathbf{G}'},$$

with, for a square rod of side $s$, $\sigma(\mathbf{G}) = (\sigma_{\mathrm{t}} - \sigma_{\mathrm{air}})\,ff\,\mathrm{sinc}(G_x s/2)\,\mathrm{sinc}(G_y s/2)$ ($\mathbf{G} \neq 0$). We use $15^2 = 225$ waves (729 waves change the edges by 0.1–0.4 %). Only modes symmetric with respect to the plane containing the sound direction count (the antisymmetric ones are not excited by a wave arriving head-on). The gap is $f_{\inf} = \max_k f_1(k)$, $f_{\sup} = \min_k f_{n\ge 2}(k)$, with $k$ from Γ to X ($|\mathbf{k}| = \pi/a$) or from Γ to M ($|\mathbf{k}| = \sqrt{2}\pi/a$). The attenuation comes from the inverse problem (given $\omega$, a complex $k$ comes out), $(k^2 M_2 + k M_1 + M_0)\,\mathbf{p} = 0$, linearised into a problem of double dimension; $\kappa$ is the smallest $|\,\mathrm{Im}\,k|$. With only the two waves $e^{\pm i\pi x/d}$, $f_{\sup} - f_{\inf} \propto |\sigma(2\pi/d)|$: the gap widens with the amount of brick per plane.

***Effective medium..*** At low frequency, $\kappa_{\mathrm{eff}} = (1-ff)\kappa$ and $\rho_{\mathrm{eff}} \simeq \rho(1+ff)/(1-ff)$ (rigid rods), hence $c_{\mathrm{eff}} = c/\sqrt{1+ff}$ and $f_{\mathrm{gap}} \approx f_B/\sqrt{1+ff} = 0.90,\ 0.93,\ 0.95,\ 0.96\,f_B$ for $a = 32$–$56\,\mathrm{mm}$.

***Complete gap..*** There is a gap along all directions if $f_{\sup}(\Gamma X) > f_{\inf}(\Gamma M)$. Along $\Gamma M$ the diagonal of the brick (22.3 mm) occupies almost the whole distance between planes ($a/\sqrt{2} = 22.6\,\mathrm{mm}$ for $a = 32\,\mathrm{mm}$): the medium looks almost uniform and the $\Gamma M$ gap is narrow, so the predicted ratio $f_{\inf}(\Gamma M)/f_{\inf}(\Gamma X)$ is 1.59, 1.51, 1.47 (and not $\sqrt{2}$) and there is no overlap for $2\times 2$ bricks ($-7\,\mathrm{Hz}$ even for $a = 24\,\mathrm{mm}$).

***Uncertainty of the depth..*** $D = T_0 - M$; with $n_{\mathrm{base}} = \log_2(0.70/0.35)/(1/6) = 6$ and $n_{\mathrm{win}} = \log_2(1.1/0.9)/(1/6) = 1.74$ independent points, and the variance of the median being $\pi/2$ times that of the mean, $\mathrm{Var}\,D = \left(\frac{\pi}{2}\frac{1}{6} + \frac{1}{1.74}\right)\sigma^2$, i.e. $\sigma_{\mathrm{rip}}(D) = 0.92 \times$ ripple. The dispersion of the references uses weights $w_k = 1/\mathrm{rip}_k^2$: $\sigma_{\mathrm{ref}}^2 = \sum_k w_k (x_k - x_0)^2 / \sum_k w_k$.

***Fits..*** For $y = mx + b$ with $w_i = 1/\sigma_i^2$: $m = (S\,S_{xy} - S_x S_y)/\Delta$, $b = (S_{xx} S_y - S_x S_{xy})/\Delta$, $\sigma_m^2 = S/\Delta$, $\sigma_b^2 = S_{xx}/\Delta$, $\Delta = S\,S_{xx} - S_x^2$ [6]; the uncertainty in $x$ enters through $\sigma_{i,\mathrm{eff}}^2 = \sigma_{y,i}^2 + (m\sigma_{x,i})^2$, minimising $\chi^2(m,b)$ numerically. If $\chi^2/\nu > 1$, $\sigma \to \sqrt{\chi^2/\nu}\,\sigma$; uncertainties are never shrunk. The regressions were checked against `numpy.polyfit`, `scipy.optimize.curve_fit` and `scipy.odr`.

## C Code

Repository: https://github.com/FM767/LFE-acoustics.

- `cristal_simples/` and `cristal_lego_g/cristal_lego_g/`: measurement apps ($\Gamma X$ and $\Gamma M$; Python, `sounddevice`) with the measurement plan, quality control and the recordings (`projeto/dados/<step>/*.wav` + `meta.json`; repeated measurements in `_antigos/`, which do not count).
- `cristal_lego_g/analise 2.0/`: joint analysis of the two directions. `analise.py` (spectra, $T(f)$, quantities, uncertainties, fits), `teoria.py` (PWE), `graficos.py`, `app.py` (interface); results in `resultados/` (`resultados.txt`, `tabela.csv`, `ajustes.csv`, `teoria.txt`). All equations, in order, in `EQUACOES.md`.
- `relatorio/`: this report. `fazer_figuras.py` generates Figs. 2–3 and Table A1 from `dados_figuras/`; `dados_figuras/curvas_T.py` recomputes the curves of Fig. 2 from the `.wav` files (same method, numpy only).

Parameters: FS = 48000, NPERSEG = 4096, band 0.5–12 kHz, segment 0.8–10.3 µs, 1/6-octave smoothing, 6 dB threshold, $\sigma_a = 0.1\,\mathrm{mm}$, $\sigma_\theta = 1\,^\circ\mathrm{C}$.

## D Use of AI

We used Claude (Anthropic) as an assistant: in planning the experiment and choosing the parameters, in writing the measurement and analysis code (which we reviewed and tested: the regressions were compared with independent libraries and the coverage of the uncertainties by simulation) and in drafting this report from the results files. The set-up, the measurements, the decisions on what to measure and the verification of the numbers against `resultados.txt` and `teoria.txt` were done by us.